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Factors and barriers for peer-to-peer carsharing adoption in Greece

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Abstract
Peer-to-peer (P2P) carsharing has become a promising alternative to traditional car ownership, offering economic, environmental and social benefits. Despite its increasing acceptance worldwide, the uptake of P2P carsharing in Greece remains limited. This study deploys a survey to identify the factors influencing the uptake of P2P carsharing in Greece and a total of 203 responses are collected. The survey examines a set of factors, including demographic characteristics, travel behavior, technological readiness, environmental attitudes and perceptions of carsharing. Both descriptive statistics and ordinal regression are used in the analysis to identify significant predictors of adoption. The results show that 60% of respondents are not aware of P2P carsharing services, 35% had heard of them but had never used them, and only 5% had experience of such services abroad. Ordinal regression models show that the prospect of financial benefits - such as compensation for maintenance and insurance costs—correlates positively with the willingness of car owners to offer their car. At the same time, renters perceive the lower costs compared to traditional rentals as the main motivating factor. Despite these opportunities, trust and safety concerns are found to be important barriers on both sides. Car owners worry about misuse and damage to the vehicle, while renters express concerns about the condition of the vehicle and the trustworthiness of unknown owners. Clear insurance terms and reliable platforms are therefore key to promoting trust in P2P transactions. Furthermore, while environmental awareness plays a minor but statistically significant role, it is overridden by economic and safety considerations when it comes to willingness to participate. The results contribute to the broader discourse on sustainable mobility by highlighting the role of local factors in the adoption of innovative transportation solutions.

Keywords Peer-to-peer carsharing, Shared mobility, Sustainable mobility, User adoption, Barriers and drivers

1 Introduction

Urban areas around the world have seen a surge in car use in recent decades [1], leading to a variety of interrelated problems, including traffic congestion [2, 3], excessive energy consumption [3, 4] and a lack of adequate parking infrastructure [5, 6]. As cities continue to expand and populations concentrate in city centers, over-reliance on private vehicles often leads to extended commuting times [7, 8], increased stress for commuters [7] and a reduction in the quality of life [9, 10]. The situation is exacerbated by inadequate public transport networks in many regions, which do not offer a convenient or efficient alternative to the private car. Consequently, these restrictions reinforce a cycle of car dependency that leads not only to congestion on roads and highways, but also to environmental damage in the form of noise pollution and poorer air quality [11–13].

In addition to these challenges, economic barriers further limit mobility for many urban residents. Affordability of transportation is a crucial factor affecting the overall accessibility, as high costs often exclude vulnerable groups from fully participating in economic and social activities [14], Guzman and Oviedo, 2018). Research consistently highlights the need for low-cost transportation options [15], emphasizing that their implementation can significantly enhance both sustainability and accessibility within transportation systems [16]. Ensuring that transportation remains affordable is essential for promoting social inclusion and equitable mobility [17].

Greek cities are an example of these general trends. They are facing serious problems due to intensive urbanization and historically poor urban planning [18]. Overcrowded streets and limited public spaces have become the norm, while parking is a recurring problem for residents and local authorities alike. Insufficient investment in public transportation infrastructure further encourages car dependency, since a limited number of attractive alternatives exist to reduce private vehicles in Greece. Along with other European countries, Greece has committed to these environmental initiatives to reduce the negative effects of pollution and to improve living standards in cities. In recent years, Sustainable Urban Mobility Plans (SUMPs) have been developed and implemented in Greek cities, which fit into the broader European paradigm of promoting clean and efficient transportation. These policy frameworks promote the use of cleaner modes of transport and the introduction of shared mobility solutions to reduce congestion and minimize environmental damage.

Among the many strategies that fall under the umbrella of shared mobility, carsharing stands out as a promising approach to alleviating traffic problems. Carsharing can reduce individual car ownership by offering the use of cars on demand without the need for permanent private ownership [19]. According to Shaheen et al. [20], carsharing gives members access to a fleet of vehicles for short-term use, significantly reducing the need to own a vehicle. Litman [21] describes carsharing as a car rental service designed to replace private vehicle ownership, with an emphasis on affordability and ease of use. Danielis et al. [22] highlight carsharing as a model of short-term car rental where users pay based on time or distance traveled, often supported by technology for booking and usage management. Additionally, carsharing has been rated as the best alternative among highway vehicles and modes in terms of sustainability performance, highlighting its potential in urban contexts [23]. Carsharing schemes have also shown considerable potential to serve as an intermediate solution between complete car dependence and total reliance on public transportation, filling a critical gap in urban mobility networks [24]. Shaheen et al. [20] identify four basic business models of carsharing: (1) business-to-consumer (B2C),(2) business-to-government (B2G); (3) business-to-business (B2B); and (4) peer-to-peer (P2P). While B2C carsharing has become an integral part of urban mobility, P2P carsharing takes the principles of the sharing economy one step further by enabling private individuals to share their private vehicles with others via digital platforms, creating a more decentralized system [25].

Among the most famous P2P sharing platforms Turo operates in the United States, Canada, the United Kingdom and Germany and offers a wide range of vehicles, from everyday cars to luxury models [26]. Getaround is another major player that serves users across the North America and Europe and has acquired the French platform Drivy in 2019 [27]. In Europe, SnappCar stands out, particularly in the Netherlands, Germany and Denmark, promoting carsharing in the community [28]. HyreCar aims specifically at ride-sharing drivers and enables them to rent vehicles for services such as Uber and Lyft [29].

In theory peer-to-peer (P2P) carsharing would be particularly attractive in providing solutions for Greece, as it allows the utilization of existing vehicle fleets. and promotes the reduction of cars on the streets. However, despite these obvious benefits, P2P carsharing initiatives in Greece have yet to achieve significant success, as both start-ups and international platforms continue to struggle with sustainable adoption rates or profitability. Ridemind, which operated from 2018 to 2020, ceased operations due to mobility restrictions imposed during the COVID-19 pandemic. The recently launched P2P carsharing service offered by Sync is still in its early stages with a very limited market share, and exact adoption figures are not available. Given the potential of P2P carsharing to address urban mobility challenges, it is important to examine the reasons behind these unsuccessful attempts, as it could pave the way for the development of more effective shared mobility solutions in the future.

Considering the above, the present study aims to investigate the factors hindering the adoption of P2P carsharing in Greece and identifying possible factors for future success. It focuses on understanding how the public perceives the idea of P2P carsharing in the absence of an existing scheme. The paper aims to address the following research questions:

(a) What is the level of awareness and general understanding of P2P carsharing among residents of Greece?

(b) What factors influence individuals' willingness to participate in a P2P carsharing scheme, either as vehicle owners or renters?

(c) What concerns and perceived barriers limit participation in P2P carsharing, and how do these concerns affect stated intentions?

The rest of the paper is organized as follows: Sect. 2 provides a comprehensive overview of the relevant literature. Section 3 describes the research methodology, detailing the approach to data collection, the survey and the analysis techniques used. Section 4 presents the results of the survey and provides a systematic examination of the key findings and their practical implications. Finally, Sect. 5 provides a combined discussion and conclusion, making policy and planning recommendations, identifying limitations and suggesting future research directions.

2 Literature review

P2P carsharing has emerged as a novel and flexible mobility option [20]. According to Hampshire & Gaites [30], "*peer-to-peer (P2P) carsharing allows car owners to convert their personal vehicles into shared cars that can be rented to other drivers on a short-term basis*". The main difference between this service and ride-sharing is that passengers are driven in a vehicle owned and operated by the owner [15]. The P2P model has gained significant attention from researchers and transport planners because it makes use of idle vehicles without the need for major investment in infrastructure [30–32]. Hence, the existing literature has examined a variety of financial, environmental and social factors that act as enablers or barriers to is adoption, indicating the complexity of the implementation and broader acceptance of P2P carsharing systems.

Research has indicated that the acceptance of P2P carsharing depends on several financial and societal factors. Younger people are more likely to participate in P2P carsharing, both as renters and providers, while men are typically more inclined to list their vehicles on P2P platforms, often motivated by economic returns and a higher tolerance for risk [33, 34]. Higher-income individuals are more likely to provide vehicles for P2P carsharing, as they often own multiple vehicles, some of which may be underutilized [34]. On the renter's side, low to middle income groups show higher interest in P2P carsharing because it is affordable compared to vehicle ownership [34]. Finally, participation is positively correlated with a higher level of education [35]. These demographic characteristics indicate that one of the most important factors that motivates both car owners and users to participate in a P2P carsharing scheme is the financial benefit. Owners use P2P carsharing to profit from underutilized vehicles and cover their ownership costs, with the additional income being especially valuable for low-income households [36, 37]. For renters, the model provides access to vehicles without the long-term financial burden and fixed costs of ownership, offering flexibile and cost-effective usage of vehicles [30, 36, 38]. A particular benefit of the model has been shown in underserved areas where traditional carsharing services are less economically viable [39]. However, despite its economic potential, the adoption of P2P carsharing could face challenges such as regulatory barriers and fragmented insurance policies, while its broader effects on transportation equity and affordability remain uncertain [34].

From an environmental perspective, P2P carsharing offers a promising way to reduce greenhouse gas emissions and improve sustainability in urban areas, where shared mobility is most essential, feasible and widespread [37]. Each shared vehicle has the potential to replace several private vehicles, contributing to a significant reduction in the overall environmental footprint of transportation. By reducing the need to own a car, P2P carsharing decreases the number of vehicles on the road, alleviating traffic congestion and air pollution [40, 41]. Also, the decreased demand for private parking spaces associated with P2P carsharing provides urban land that can be used for green spaces, pedestrian walkways or other community needs [30], reinforcing the process of sustainable urban planning and enhanced use of the available resources. This can facilitate behavioral changes that promote sustainability, influencing many participants to reduce their dependence on private vehicles and opt for public transportation, walking or cycling instead [42]. Many participants could be motivated by the opportunity to contribute to sustainability goals [30, 39, 43]. Another environmental parameter that is associated with P2P carsharing is energy. The model supports the adoption of alternative energy vehicles, while some platforms incentivize or prioritize electric and hybrid cars, helping to further reduce emissions and dependence on fossil fuels [41]. Furthermore, by using existing vehicles more efficiently, P2P carsharing can extend the life cycle of cars, reducing the environmental impact associated with manufacturing and disposal [40].

Cultural values also influence participation in peer-to-peer carsharing, with collectivist societies showing a higher willingness to participate due to norms of sharing and mutual support [42] 8) . P2P carsharing is considered to strengthen community bonds, as sharing vehicles promotes trust and cooperation in the neighborhood and provides mobility options for people who would otherwise not have access to transportation [36] (Wilhelms et al., 2017), whether due to affordability or by choice not to own a car. This accessibility promotes inclusion and community equality, especially in areas, mostly rural where public transport is inadequate or inconsistent [36, 44, 45] . Studies highlight the role of digital platforms in enabling new forms of interaction, trust-building and collaboration between participants, thereby strengthening community bonds [37, 38].

Flexibility is another key factor influencing the participation in P2P carsharing. Car owners often participate because they can choose when and how their car is rented, so their schedule is only minimally disrupted [33]. However, psychological barriers such as anticipated stress and fear of vehicle misuse can discourage participation. Studies show that these factors, combined with a general discomfort with sharing personal goods, limit acceptance [46, 47].

P2P carsharing platforms mainly function through digital applications and websites. However, the existing literature suggests that concerns remain regarding the users' digital readiness and their ability to interact effectively with platform features These concerns extend to the perceived safety and reliability of digital tools, which can influence participation. For example, concerns include the fear of imprecise information, as most platforms rely extensively on user feedback systems to maintain security and service quality. Ethical issues, such as trust, surveillance, transparency, consent, and control over privacy, personal data ownership and social discrimination raise additional skepticism toward the digital platforms [15]. Moreover, there is evidence that elderly individuals, people with low levels of education, and those with physical or cognitive difficulties often encounter challenges in using emerging technologies and data-driven transport solutions [15, 48]. Advancements in software and algorithms are expected to mitigate these challenges, developing inclusive and user-friendly platforms and services adapted to each of the users' needs. Enhanced recognition of user preferences and dynamic processes of ride-matching are important for tailored services aiming to address the diverse user needs [15]. Trust mechanisms such as ratings, reviews and insurance guarantees reduce perceived risks and encourage both owners and tenants to participate [38]. Platform reliability and technological advancement play a critical role in allaying concerns about vehicle safety and liability, which are often barriers to adoption [20].

The extent of the benefits of P2P carsharing depends on user behavior and overall usage patterns. When carsharing leads to an increase in car use by users who previously relied on non-motorized or public transport, the net environmental impact may be reduced. This underlines the importance of integrating P2P carsharing into multimodal, sustainable urban transportation strategies [39]. P2P carsharing has also been linked to changing consumer behavior and attitudes towards ownership. By encouraging access rather than ownership, a cultural shift away from traditional car dependency is encouraged, particularly among younger generations who favor flexibility and sustainability [38, 39]. However, it is important to note that the growth of the P2P carsharing industry and the increasing contribution of social media in promoting carsharing service, significant privacy issues will arise [31].

Despite the increasing attention given to shared mobility in many parts of the world, comparatively little empirical work has addressed the particular circumstances of the Greek market. One of the few studies that have looked at carsharing in Greece to date is that of Vassi et al. [48], who used qualitative methods - specifically semi-structured interviews - to investigate motivational factors and personal characteristics of potential carsharing users. Their findings emphasize the role of familiarity with the concept, convenience, attitude, daily habits, ease of use and economic considerations in people's willingness to engage in carsharing. While these findings provide a valuable first insight into the broader factors influencing the adoption of carsharing, there is no specific research on P2P carsharing in Greece to date. This significant gap underscores the urgent need for more focused research into consumer perceptions in the country's unique socio-cultural and infrastructural context.

3 Methodology

3.1 Research design

This study uses a quantitative research approach to investigate the factors influencing the adoption of P2P carsharing in Greece. The primary data for this study was collected using a structured online questionnaire distributed between April and May 2024. The questionnaire was designed based on a comprehensive review of the literature on shared mobility, carsharing and consumer behavior theories. The survey instrument included closed-ended Likert scale questions to assess respondents' attitudes, perceptions and willingness to participate in P2P carsharing. Participants were briefly informed about the operation of P2P carsharing to provide a basic understanding of its concept. To maximize participation and data quality, the survey was conducted online through a secure platform, allowing access from various digital devices, and required about seven minutes to complete. Prior to distribution, a pilot study was conducted with ten respondents to improve question clarity, wording and overall coherence. Based on the feedback, adjustments were made to minimize response bias and increase the reliability of the data collected. In addition, measures such as randomizing the order of questions and using neutral wording were taken to mitigate potential response bias.

3.2 Sampling strategy

A snowball sampling was used to recruit participants who met certain eligibility criteria. As there is no central database of potential carsharing users in Greece, this approach allowed for efficient recruitment through existing social and professional networks. The sample comprised of 203 respondents, all of whom being permanent residents of Greece, aged between 18 and 65 and living in urban areas. The recruitment process was initiated through digital platforms and academic networks, where initial participants were asked to share the survey with others who met the study's criteria. This method allowed access to a diverse yet relevant population while taking into account practical constraints such as lack of time and resources. Although the snowball sampling introduces potential biases, these concerns were mitigated by maximizing the sample size and ensuring demographic diversity in the final dataset.

3.3 Survey instrument

The questionnaire was designed to comprehensively cover the most important aspects that influence the acceptance of P2P carsharing. An introductory note at the beginning of the questionnaire provided respondents with a short explanation of the concept of P2P carsharing, outlining how such platforms operate, the roles of vehicle owners and renters, and the basic principles of short-term peer-to-peer vehicle access. This ensured that all participants shared a common understanding of the service before answering the relevant items. The survey was divided into five thematic sections. The first section looked at demographic characteristics, including age, gender, education level, employment status, income level and household composition to create a socio-economic profile of respondents. The second section examined travel behavior and car ownership, identifying commuting habits, main modes of transportation and accessibility to private vehicles. The third section explored respondents' familiarity with smart mobility technologies and environmental awareness, assessing their use of digital applications for transportation and their attitudes towards sustainable mobility. The fourth section examined willingness to participate in P2P carsharing, distinguishing between potential users as car owners willing to rent their vehicle and those interested in renting a vehicle through a carsharing platform. The final section explored perceived barriers and incentives to participation, addressing concerns such as trust issues, cost factors, safety and insurance considerations, and potential economic and convenience benefits. The questionnaire included a combination of multiple-choice questions and Likert scale ratings to allow for comprehensive statistical analysis while ensuring clarity and simplicity of responses for participants.

3.4 Data preparation and analysis

In order to ensure completeness of the data, all survey items were mandatory and only fully answered questionnaires could be submitted. As a result, the dataset contained no partial responses, and no additional data cleaning procedures were required. The collected survey responses were analyzed using descriptive and inferential statistical methods in SPSS 29 and Microsoft Excel. Descriptive statistics, including frequency distributions, means, and standard deviations, were used to summarize demographic characteristics, travel behavior, and attitudinal responses. To assess the internal consistency of the attitudinal items included in the questionnaire, a Cronbach's alpha reliability analysis was conducted on the set of Likert-scale items capturing perceptions, motivations, and concerns related to peer-to-peer carsharing. The resulting Cronbach's alpha value was 0.875, indicating high internal consistency.

In addition to the descriptive analysis, two statistical models are developed to quantify and understand the relationship between respondents' willingness to adopt P2P carsharing as car owners or renters and various survey-based factors that could explain or predict this willingness. Since the degree of willingness was measured with an ordered, categorical dependent variable (e.g. from "strongly disagree" to "strongly agree"), the method of ordinal regression was selected. This approach is particularly well suited to analyzing results ordered on an ordinal scale, and it allowed the identification of how closely and reliably different explanatory variables, such as perceived benefits or concerns, were associated with respondents' willingness to participate in P2P carsharing.

Ordinal regression model evaluation should go beyond examining coefficient signs and significance levels [49], as these practices overlook the characteristics of ordinal logit models, whose coefficients are not easily interpreted because the models are inherently non-linear and do not provide effect sizes in the usual sense [50]. Parady et al. [51] recommend using elasticities and marginal effects, which provide a clearer understanding of how explanatory variables affect the probabilities of specific outcome categories. This study calculated elasticities for all independent variables and derived marginal effects using R language, as equivalent functions are not available in SPSS.

3.5 Sample description

A detailed description of the sample is provided in Appendix B. The majority of participants were female (53.2%), while male participants made up a lower percentage (46.8%). This distribution aligns with the 2021 Greek census, in which women represented 51.7% of the population and men 48.3%. Regarding age, there was a relative balance between the 26–35 age group (23.2%) and the 46–55 age group (24.1%). According to the 2021 census [52], individuals aged 26 to 35 comprised about 11% of the national population, while those aged 46 to 55 accounted for roughly 15%.

Additionally, regarding educational attainment, the majority were university graduates, with or without postgraduate or doctoral qualifications (77.8%). This figure contrasts sharply with the educational profile of the general population, where only about 27% hold a university degree or higher, indicating that the sample is considerably more educated than the national average. Furthermore, in terms of the monthly income profile, the majority earns between €1,500 and €2,500 (28.1%), followed by the €1,001–€1,500 group (19.2%).

Regarding the professional status of the research participants, the majority were private sector employees (37.9%), followed by public sector employees and self-employed professionals. Concerning household composition, the largest group comprised four-member families (39.4%), followed by three-member families (22.2%). Furthermore, an overwhelming majority possessed a private vehicle (> 90%). In addition, people with driver's license constituted the vast majority of the sample (87.7%).

Regarding the frequency of access to any of the household vehicles among the participants, the majority reported having access "always" (67%). Moreover, when asked if they were familiar with P2P carsharing, the majority responded negatively (59.6%), and 35% of those being aware they had never used it.

These demographics indicate a sample of middle- to higher-income professionals, which suggests that the adoption of P2P carsharing could be facilitated by their digital readiness and car ownership rates [33, 34]. On the other hand, their unfamiliarity with P2P carsharing might raise their concerns about trust, liability, and vehicle security can affect their actual participation.

3.6 Ethical considerations

The study complied with ethical guidelines and ensured voluntary participation, anonymity and confidentiality of data in accordance with the EU GDPR regulation (EU 2016/679). Informed consent was obtained from all participants, and no personally identifiable information was collected. Furthermore, the study was conducted in accordance with the ethical principles and approval protocols of the Ethics Committee for Research of the National Technical University of Athens (NTUA).

4 Results

4.1 Travel behavior and factors in mode choice

Respondents were asked to describe the main purpose of their typical daily trip, indicate the mode they usually use, and specify the usual trip duration. The questionnaire also included a separate item on the number of trips made during a typical day, with outward and return legs counted as two trips. Together, these items provide an overview of respondents' everyday mobility patterns. The variables reported in this subsection are descriptive and are included to provide contextual background on travel behavior, supporting the interpretation of the results presented in the following sections.

The analysis shows that the predominant motive for the main purpose of daily trips among the respondents is work or education, with 81.8% of participants stating that their daily trips involve one of these two activities (Table 1). This finding underscores the central importance of commuting for work or academic purposes in the daily mobility behavior of the sample population. In contrast, domestic commitments accounted for 8.4% of responses. In addition, a smaller proportion of respondents cited shopping (4.9%) and leisure (4.4%) as primary travel purposes, and an insignificant proportion (0.5%) cited other reasons.

A clear preference for private transport emerges based on the responses (see Fig. 1). Most participants travel as drivers of their own cars (60.1%), while only small shares rely on motorcycles (5.4%) or travel as passengers in private vehicles (3.0%). Public transport accounts for 28.1% of main trips, and walking remains limited (3.4%). This pattern indicates a strong dependence on private motorized travel within the sample.

Nearly half of the respondents complete three to four trips per day of medium duration, indicating a mobility pattern where carsharing could serve routine needs. Multimodal behavior is present in about two-fifths of the sample, and more than half are accustomed to walking reasonable distances, both of which suggest some openness to shared options. At the same time, public transport is often seen as insufficient.

Travel duration and safety emerge as the strongest factors in mode choice, followed by flexibility and trip purpose (see Table 2). Comfort and affordability are moderately important, while privacy receives less emphasis. Environmental friendliness is the most divisive factor, with opinions split between those who value it highly and those who view it as unimportant.

The high reliance on private vehicles, combined with the importance of flexibility, comfort, and travel duration, may positively influence participation in P2P carsharing. The relatively low use of public transport suggests potential dissatisfaction with this

Table 1 Trip purpose

	Frequency	%
Work/ Education	**166**	**81.8**
Household Obligations	17	8,4
Shopping	10	4,9
Leisure Activities	9	4,4
Other	1	0.5
Total	203	100,0

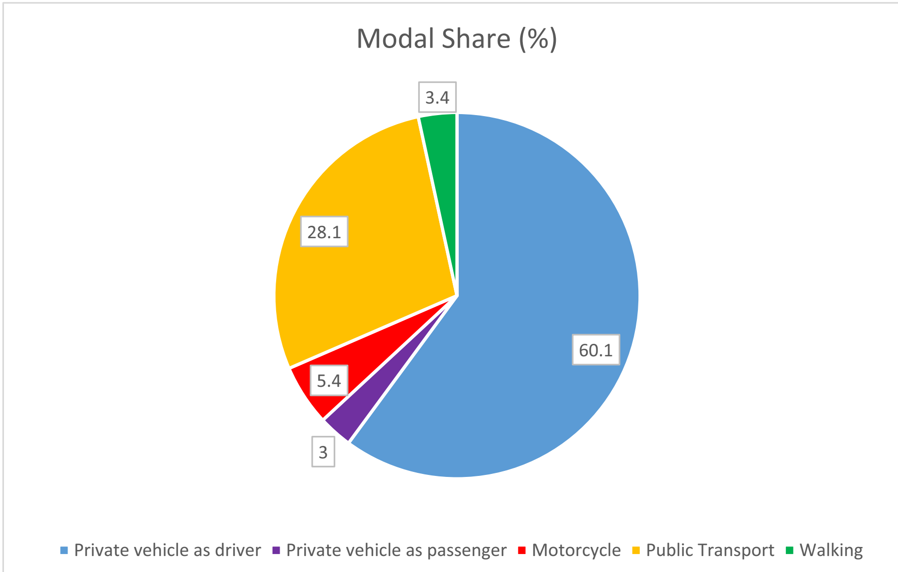

Fig. 1 Modal share

Table 2 Factors affecting mode choice (when 1 being "Not important at all" and 5 being "Very important")

	Purpose of travel	Duration of travel	Comfort	Privacy	Safety	Flexibility	Environ-mentally friendly	Af-ford-ability
Mean	3.67	4	3.57	3.07	3.89	3.77	2.84	3.51
Stan-dard Devia-tion	1.123	1.027	1.029	1.221	1.116	0.97	1.162	1.105

Table 3 Use of the equipment or applications for trip planning/mode choice (1 being "Never" and 5 being "Always")

	Computer or laptop	Smartphone or tablet	Google Maps	Telematics	Taxi apps
Mean	2.25	3.78	3.98	2.45	2.09
Standard Deviation	1.172	1.158	0.850	1.313	0.913

mode, which P2P carsharing could help address by offering a more convenient and reliable alternative .

4.2 Technology awareness and factors influencing carsharing adoption

Respondents rely heavily on smartphones and Google Maps for planning their trips, with roughly 69% reporting frequent or always use of smartphones (ratings 4 or 5) and about 79% doing the same for Google Maps. By contrast, around 68% use computers or laptops rarely or never (ratings 1 or 2), while telematics and taxi apps see similarly low uptake—over 57% and 70%, respectively, report rarely or never using these options (see Table 3).

Among the factors shaping carsharing adoption, economic benefit and vehicle cleanliness rank highest, followed by free parking, vehicle availability, and ease of use (see

Table 4 Importance of factors for using carsharing (1 being "Not important" at all and 5 being "Very important")

	Possibility of renting different types of vehicles	Possibility of renting electric vehicles	Comfort of shared vehicle	Free parking spaces for shared vehicles	Availability of shared vehicles	Ease of use	Environmentally friendly	Economic Benefit	Cleanliness inside the vehicle
Mean	3.05	2.69	3.44	3.87	3.82	3.70	3.23	3.87	3.93
Standard Deviation	1.189	1.171	1.145	1.172	1.141	1.146	1.123	1.105	1.071

Table 5 Perceptions of users about car ownership (1 being "Strongly disagree" and 5 "Strongly agree")

	It is necessary to own a car	Owning a car is a matter of social status	Owning a car offers freedom and flexibility	I know exactly what the annual cost of owning & maintaining a car is
Mean	3.53	2,10	4.20	3.60
Standard Deviation	1.272	1.060	0.936	1.149

Table 4). Comfort and environmental friendliness are of moderate importance, while variety of vehicle types and access to electric vehicles are less influential.

Respondents generally view car ownership as necessary and are aware of its costs, though they do not associate it with social status (see Table 5). Instead, ownership is strongly linked to freedom and flexibility, which remains the dominant motivation. These findings suggest that car ownership is strongly correlated with perceptions of convenience, ease of use, and reliability, which may constrain the adoption potential of P2P services. When individuals consider car ownership as essential, they are less likely to alter their mobility habits toward alternative modes of transportation such as P2P.

These results indicate that there is sufficient digital readiness to support the use of app-based carsharing platforms, although greater efforts may be needed to familiarize users with tools such as telematics. Also, while the benefits of carsharing are recognized, car ownership still appears to hold greater importance for users. The variables discussed in this subsection are considered as potential explanatory factors. Selected attitudinal and perception-related items are subsequently included as independent variables in the ordinal regression models, based on their relevance and statistical significance during model development (see Sect. 4.4.).

4.3 Willingness to participate in P2P carsharing

This section describes how respondents assessed their likelihood of participating in a P2P carsharing scheme, either as owners or renters, which constitutes the dependent variable in the ordinal regression models. Car owners were asked to rate their willingness to rent out their private vehicles through a P2P platform. They also evaluated various possible motivations, such as financial gain or environmental benefits, and concerns including trust in other users, vehicle wear, insurance coverage, and privacy. Similarly, potential renters were asked about their willingness to rent a privately owned car and to

Table 6 Users' concerns for participation as car owners (1 being "Not important" and 5 being "Very important")

	Lack of trust in other users	Vehicle wear and tear	Insurance and liability	Maintenance and mechanical breakdowns	Cleanliness of the vehicle on return	Privacy & data protection issues
Means	4.05	4.03	3.98	4.07	3.85	3.64
Standard Deviation	0.976	0.980	0.951	0.912	1.025	1.114

Table 7 Barriers to participation as renters (1 being "Not important" and 5 being "Very important")

	Lack of vehicle availability	Feeling unsafe about the condition of the vehicle	Feeling unsafe towards the owner of the vehicle	Lack of familiarity with the service	Complexity of the hiring process
Mean	3.64	3.92	3.72	3.31	3.38
Standard Deviation	1.101	0.994	1.082	1.222	1.156

assess factors that might encourage or discourage them, such as cost, immediate availability, safety, familiarity with the service, and the complexity of the rental process.

Among car owners, economic benefit is by far the strongest motivator for participating in a P2P carsharing scheme: about 70% rate it as important or very important. Environmental concerns, though still relevant, rank lower, garnering 39% in top ratings. On the other hand, lack of trust in other users (approximately 77% rating important or very important) stands out as the most prominent worry, closely followed by vehicle wear and tear and maintenance/ mechanical breakdown (each exceeding 70% in top concern ratings). In general, issues related to insurance and liability, cleanliness, and privacy also register as concerns, though to a slightly lesser extent (see Table 6).

Provision of full vehicle insurance emerges as the dominant factor for owners considering P2P carsharing: nearly 68.5% of respondents rate it as very important and 19.7% rate it as important. Close behind is flexibility in determining vehicle availability (about 51.2% "very important," 31.5% "important") and the ability to approve renters (about 48.3% "very important," 30.5% "important"). Owners also value setting their own rental price (around 39% "very important," 33% "important") and automated vehicle handovers (approximately 35% "very important," 37% "important"). Despite these preferences, willingness to actually rent out one's car is relatively low: fewer than 25% express a clear inclination (ratings 4 or 5), while nearly 47% lean negative.

From a renter's perspective (see Table 7), lower costs compared to traditional rentals (about 43% very important, 35% important) and immediate vehicle availability (roughly 35% very important, 39% important) are key motivations. Concerns mainly center on safety (over 70% worry about vehicle condition) and lack of vehicle availability (nearly 59% rate this a top concern), alongside uncertainty about feeling unsafe toward the owner (about 65% rate it 4 or 5). Moreover, a significant majority (over 80% in total) emphasize full insurance and transparent terms of use as prerequisites for renting from a private individual. Overall, while cost savings and convenience can drive interest in P2P carsharing, apprehensions about safety, trust, and familiarity with the service remain significant barriers to widespread adoption.

These findings regarding both owner's and renter's perspective highlight that while the economic appeal of carsharing is important, a wider participation of participants highly depends on addressing concerns around trust, risk and vehicle security.

A clearer picture of stated willingness emerges when examining the distribution of responses across the five-point scale (see Fig. 2). Among car owners, more than half fall into the two lowest categories, with 26.6% stating they would not participate at all and 20.7% expressing little willingness. A smaller group reports moderate interest (29.1%), while those indicating they would participate "much" or "very much" remain limited (16.3% and 7.4%, respectively). The pattern differs among potential renters, who are generally more open to the service. Only 10.8% state no willingness, and 12.3% indicate little willingness. The largest share falls in the middle category, with 32% selecting an average level of willingness, and a similar number (33%) expressing a strong inclination to use such a service. Additionally, 11.8% report very high willingness. This contrast shows that renters are substantially more receptive to P2P carsharing than owners, while owners remain cautious and predominantly reluctant.

4.4 Model results

The development of both ordinal regression models followed a structured sequence of screening, specification, and diagnostic assessment before substantive interpretation. All variables collected through the survey were examined in an initial exploratory stage to identify their potential contribution. This stage included demographic characteristics such as age, gender, education, income, and household composition, as well as indicators of mobility behavior, including main mode of transport, trip purpose, frequency of daily travel, and access to private vehicles. These variables were entered individually and in thematic groups in preliminary model specifications. None reached statistical significance at conventional levels, and several produced unstable coefficient signs across specifications. They were therefore excluded from the final models to ensure a parsimonious structure and limit redundancy between predictors.

Attitudinal variables concerning economic considerations, environmental awareness, perceptions of trust and safety, and factors related to platform operation were

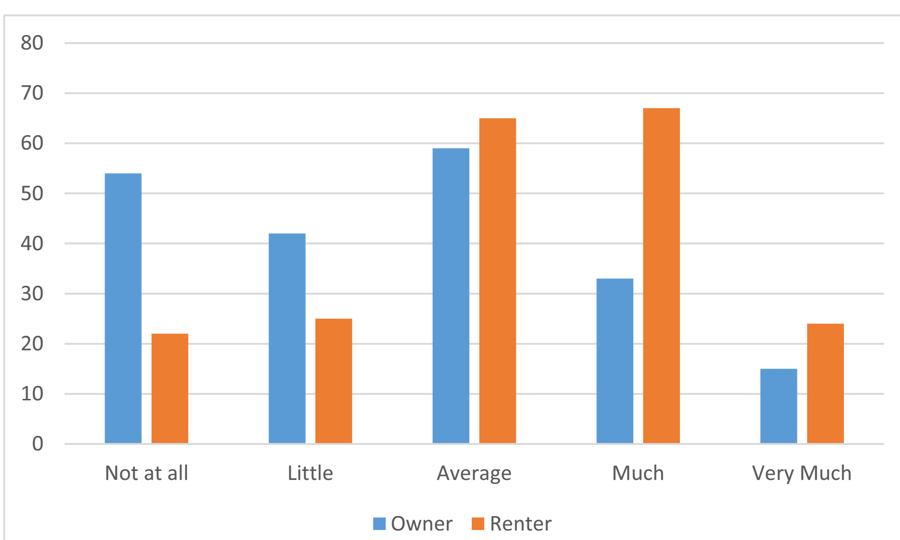

Fig. 2 Willingness to participate in a P2P carsharing scheme both as renter and owner

subsequently evaluated. Predictors were incorporated in blocks reflecting thematic categories, and each specification was evaluated using likelihood ratio tests, Wald statistics, and changes in the log-likelihood. The proportional odds assumption underlying the ordinal regression framework was tested and not violated in either model. Only after confirming adequate model fit and specification were threshold and location parameters interpreted. This procedure resulted in two models that reflect the most stable and statistically meaningful predictors of willingness to participate in a P2P carsharing scheme.

4.4.1 Model 1: willingness to participate as car owner

The overall adequacy of the ordinal regression model for car owners was first evaluated using model fitting and goodness-of-fit statistics. The −2 Log Likelihood (−2LL) decreased from 481.958 in the intercept-only model to 405.181 in the final specification. This reduction corresponds to a statistically significant likelihood ratio test ($\chi 2 = 76.777$, $df = 4$, $p < 0.001$), indicating that including the explanatory variables significantly improves model fit compared to the intercept-only model.

Model fit was further assessed using Pearson and deviance goodness-of-fit tests. The Pearson chi-square statistic was 396.233 ($df = 440$, $p = 0.934$), and the deviance chi-square statistic was 320.777 ($df = 440$, $p = 1.000$). The non-significant results of both tests indicate no evidence of model misfit, supporting the adequacy of the specified ordinal regression model.

In addition, pseudo R-squared statistics were computed to provide an estimate of the proportion of variability explained by the predictors. Cox and Snell's pseudo R-squared was 0.315, while Nagelkerke's pseudo R-squared was slightly higher at 0.331. The test of parallel lines produced a chi-square value of 14.195 ($df = 12$) and a significance level of 0.288. The non-significant result confirms that the proportional odds assumption is not violated .

The analysis classified five response categories to the dependent variable, where 1 stands for "Not at all willing" and 5 for "Very willing" to rent their own car to others via a P2P carsharing service. Four threshold parameters were estimated to delineate the latent continuum between these categories. The first threshold (Estimate = −1.301; $p = 0.107$) distinguishes between ratings 1 and 2, while the second threshold (Estimate = −0.116; $p = 0.885$) distinguishes between ratings 2 and 3. The third threshold (Estimate = 1.461; $p = 0.070$) almost reaches the conventional significance level and indicates a borderline distinction between categories 3 and 4, while the fourth threshold (Estimate = 3.036; $p = 0.000$) is statistically significant and clearly marks the transition from "somewhat willing" to "very willing" (see Table 8).

Four independent variables were included in the model to assess car owners' motivations and concerns about participating in a P2P carsharing scheme. The context of the first independent variable referred to the respondents' assessment of the economic benefit. The more car owners perceive an economic benefit in renting out their vehicle (Estimate = 0.702; $p < 0.001$), the more likely they are to indicate a higher willingness to participate. This result indicates that financial incentives play a decisive role in motivating car owners to participate in carsharing.

The second independent variable addressed environmental concerns. The results show that a higher level of environmental awareness (Estimate = 0.342; $p = 0.002$) is associated with a higher probability of a more favorable evaluation of willingness. In other words,

Table 8 Ordinal regression model analyzing willingness of users to participate in a P2P carsharing service as car owners

		Estimate	Std. error	Wald	df	Sig
Threshold	[How willing would you be to rent your own car to others via a P2P carsharing service? = 1]	−1.301	0.807	2.598	1	0.107
	[How willing would you be to rent your own car to others via a P2P carsharing service? = 2]	−0.116	0.805	0.021	1	0.885
	[How willing would you be to rent your own car to others via a P2P carsharing service? = 3]	1.461	0.805	3.292	1	0.070
	[How willing would you be to rent your own car to others via a P2P carsharing service? = 4]	3.036	0.827	13.467	1	0.000
Location	What would motivate you as a car owner to participate in a carsharing scheme between private individuals? **[Economic benefit]**	0.702	0.128	29.944	1	0.000
	What would motivate you as a car owner to participate in a carsharing scheme between private individuals? **[Environmental concerns]**	0.342	0.112	9.357	1	0.002
	What are your main concerns about your participation? **[Vehicle wear and tear]**	−0.516	0.147	12.369	1	0.000

car owners who value environmental benefits are more likely to rent their car, suggesting that sustainable mobility is an attractive proposition.

The third independent variable focused on concern about wear and tear on the vehicle. The regression model revealed a negative relationship (Estimate = –0.516; $p < 0.001$), meaning that increased concern about possible damage to one's vehicle significantly reduces the willingness to rent it out. This result underlines the influence of perceived risks associated with vehicle maintenance on owners' participation decisions.

The fourth independent variable examined the importance of obtaining approval from renter. Here, a negative coefficient (Estimate = –0.403; $p = 0.004$) indicates that the more important car owners consider the requirement of the renter's consent to be, the less willing they are to participate in a carsharing agreement. This result could reflect car owners' desire to retain control over who accesses their vehicle and thus reduce their willingness to participate if such approval is perceived as a significant obstacle.

The marginal effects for the car-owner model indicate how changes in perceived benefits and risks affect the probability of expressing greater willingness participate in a P2P carsharing service (see Appendix D). Economic benefit has the strongest positive effect, with the probability of moving to a higher willingness category increasing by about 0.71 when owners shift from "not at all important" to "unimportant." The effect remains positive in subsequent transitions, though it decreases to approximately 0.23 in the "neutral to important" range and about 0.19 when moving from "unimportant" to "neutral." Environmental concern shows smaller and less consistent effects, rising by about 0.12 in the lower transitions and declining to roughly 0.06 in the highest, indicating that environmental motivations play a secondary role compared to financial incentives. The negative marginal effects associated with perceived risks are more pronounced. Concerns about vehicle wear and tear reduce the probability of higher willingness by roughly – 0.25 in the "unimportant to neutral" transition and by about – 0.16 in the "important to very important" range, while the lowest category shows a smaller but still meaningful decline of about – 0.23. A similar pattern appears for the importance assigned to approving the renter. In the "unimportant to neutral" transition, willingness decreases by about – 0.21, followed by declines of about – 0.13 across the middle and upper transitions.

The elasticity estimates for the car-owner model indicate how sensitive willingness is to proportional changes in perceived incentives and concerns (see Appendix C). Economic benefit shows relatively modest elasticities in the lower transitions, rising from about 0.01 to approximately 0.28 and 0.23, but becomes more influential in the highest transition, where the elasticity reaches about 0.52. Environmental considerations show a steadier pattern, with elasticities clustering between about 0.35 and 0.58 across all transitions, indicating moderate and consistent sensitivity to changes in environmental perceptions. In contrast, variables reflecting risk perceptions exhibit much larger elasticities. Concerns about vehicle wear and tear yield values as high as 6.96 and 9.86 in the first two transitions, followed by smaller but still notable values of about 1.78 and 3.16 in the higher transitions. The importance assigned to approving the renter shows a similar pattern, with elasticities of about 2.15 in the lowest transition and almost 6.68 when moving from "unimportant" to "neutral," then stabilizing around 2.58 to 2.59 in the upper categories.

4.4.2 Model 2: willingness to participate as renter

The renters' model also demonstrates a statistically significant improvement over the intercept-only specification. The – 2 Log Likelihood decreased from 559.189 to 499.134, yielding a likelihood ratio statistic of $\chi2 = 60.055$ (df = 5, $p < 0.001$), confirming that the included predictors significantly enhance explanatory capacity.

Goodness-of-fit statistics further support the adequacy of the specification. The Pearson chi-square value was 653.492 (df = 651, p = 0.465), and the deviance statistic was 466.826 (df = 651, p = 1.000), with both tests indicating no evidence of model misfit.

The pseudo R2 measures suggest moderate explanatory strength, with Cox and Snell's R2 equal to 0.256 and Nagelkerke's R2 equal to 0.270. Finally, the proportional odds assumption was satisfied, as indicated by a non-significant test of parallel lines ($\chi2 = 18.027$, df = 15, p = 0.261) .

The analysis classified the response scale into five ordered categories for the dependent variable, where 1 corresponds to "Not at all willing" and 5 to "Very much willing". In the context of the ordinal regression model, four thresholds have been estimated that demarcate the latent continuum separating these five response levels. The thresholds represent the latent cut-points separating adjacent willingness categories. In this study, the first threshold (estimate = –0.509; p = 0.494) separates respondents who would score 1 from those who would score 2, while the second threshold (estimate = 0.600; p = 0.419) distinguishes between ratings of 2 and 3. The third (estimate = 2.344; p = 0.002) and fourth thresholds (estimate = 4.431; $p < 0.001$) are statistically significant, thereby signaling that the transitions from moderate to high willingness—specifically between categories 3 and 4 and between 4 and 5—are distinctly demarcated on the underlying latent scale (see Table 9).

Five independent variables were included in the model to assess various motivational and deterrence factors influencing individuals' willingness to rent a car via a P2P carsharing service.

The first independent variable captures respondents' appraisal of the immediate availability of vehicles as a motivator for participating as renters. The regression model indicates that an increase in the perceived importance of immediate availability is associated with a significant increase in willingness, as evidenced by a positive coefficient

Table 9 Ordinal regression model analyzing willingness of users to participate in a P2P carsharing service as renters

		Estimate	Std. error	Wald	Df	Sig
Threshold	[How willing would you be to rent a car from a private individual via a P2P carsharing service? = 1]	−0.509	0.745	0.467	1	0.494
	[How willing would you be to rent a car from a private individual via a P2P carsharing service? = 2]	0.600	0.744	0.652	1	0.419
	[How willing would you be to rent a car from a private individual via a P2P carsharing service? = 3]	2.344	0.765	9.396	1	0.002
	[How willing would you be to rent a car from a private individual via a P2P carsharing service? = 4]	4.431	0.795	31.042	1	0.000
Location	What would motivate you to participate as a renter in a P2P carsharing scheme? [**Immediate availability of vehicles**]	0.654	0.143	21.013	1	0.000
	What would motivate you to participate as a renter in a P2P carsharing scheme? [**Environmental concerns**]	0.291	0.118	6.088	1	0.014
	What are your main concerns about your participation? [**Feeling unsafe towards the owner of the vehicle**]	−0.362	0.132	7.567	1	0.006
	What are your main concerns about your participation? [**Complexity of the hiring process**]	−0.364	0.124	8.613	1	0.003
	State how important are the following factors for using carsharing? [**Economic benefit**]	0.290	0.129	5.044	1	0.025

(estimate = 0.654; $p < 0.001$). For each one-unit increment in the perception of immediate availability, the log odds of being positioned in a higher willingness category (i.e., moving closer to "Very much willing") are increased.

The second independent variable gauges the extent to which environmental concerns serve as a motivating factor. The model demonstrates that respondents who place greater importance on environmental issues are significantly more likely to exhibit higher willingness ratings, with a positive estimate of 0.291 ($p = 0.014$). This finding suggests that an enhanced awareness of environmental benefits may lead individuals to favor P2P carsharing as a sustainable mobility alternative.

The third independent variable reflects respondents' safety concerns, specifically their feelings of unsafety in interactions with vehicle owners. The regression coefficient for this variable is negative (estimate = –0.362; $p = 0.006$), indicating that heightened safety concerns diminish the probability of being classified in a higher willingness category. Essentially, as respondents' apprehension regarding personal safety increases, they are more inclined to report lower levels of willingness to participate in the scheme.

The fourth independent variable assesses perceptions of the procedural complexity involved in renting a vehicle. The negative coefficient (estimate = –0.364; $p = 0.003$) signifies that the more complex the hiring process is perceived, the odds of reporting higher willingness to engage in P2P carsharing decrease, which underscores the importance of user-friendly, streamlined processes in enhancing participation rates.

Lastly, the fifth independent variable examines the perceived economic benefit of using carsharing services. The positive association (estimate = 0.290; $p = 0.025$) indicates that respondents who recognize financial advantages are statistically more likely to express higher willingness to participate. Although this effect is more modest compared to the influence of immediate availability, it nonetheless highlights the role of economic considerations in the decision-making process.

The marginal effects clarify how specific attitudes influence renters' willingness to participate in a P2P carsharing service by indicating the direction and magnitude of probability shifts associated with each factor (see Appendix D). Immediate vehicle availability has the strongest positive effect, increasing the probability of selecting a higher willingness category by about 0.17 when respondents move from "unimportant" to "neutral," and by about 0.10 when moving from "neutral" to "important." Economic benefit also raises willingness, with an estimated change of about 0.40 in the lowest category, while environmental concern has a more moderate effect of about 0.10 in the "unimportant to neutral" range. In contrast, variables related to trust and procedural clarity show consistent negative marginal effects. Feeling unsafe toward the vehicle owner lowers the probability of higher willingness by about 0.07 to 0.09 across the middle and upper transitions, and concerns about the hiring process reduce it by about 0.09 to 0.14 depending on the category.

Economic benefit shows moderate elasticities across most transitions, increasing from about 0.06 in the lowest category to roughly 1.25 when respondents move from "important" to "very important," indicating that willingness responds more strongly as economic considerations become more notable. Immediate vehicle availability exhibits a steeper pattern, rising from about 0.05 in the first transition to more than 0.60 in the highest, suggesting that perceived convenience gains have a progressively stronger influence on willingness (see Appendix D). Environmental concerns yield elasticities between about 0.25 and 0.86, implying a noticeable but secondary sensitivity compared with economic and availability-related factors. In contrast, the largest elasticities are associated with perceived risks. Feeling unsafe toward the vehicle owner produces values between about 1.38 and 3.59, while concerns about the hiring process yield even higher elasticities, reaching more than 7.20 in the lowest transition.

5 Discussion and conclusions

This study investigated the factors that influence the acceptance of P2P carsharing in Greece. It examined both the willingness of car owners to rent out their private vehicles and the willingness of potential users to rent a privately owned vehicle. The survey results show that although there is a high dependence on private cars in Greece, as reflected in the high car ownership rates reported by Greek census data [52], a significant share of respondents are also interested in alternative forms of mobility, especially when issues of trust, convenience, and economic benefit are addressed. P2P carsharing, though, is not yet widely known or used by respondents, indicating that awareness and understanding of this option remain limited. These findings suggest that policies or pilot programs to increase awareness of carsharing could be considered, especially if tailored to current market conditions and prevailing mobility needs.

The results underline three important aspects. Firstly, economic considerations are a key factor for both potential owners and potential renters. Owners consistently rated financial benefits, particularly those related to maintenance, insurance, and depreciation costs, as an important motivating factor. Renters placed a high value on affordability and reported greater willingness to use P2P carsharing when they perceived it as less costly than traditional car rental services. These findings confirm previous research showing financial benefits as the main motivator for shared mobility [36]. Consequently, policy discussions on P2P carsharing adoption may benefit from considering financial incentives and regulatory frameworks that address fixed vehicle costs for owners and users.

Secondly, trust and safety concerns are the biggest barriers to adoption. Owners were particularly concerned about wear and tear of their vehicles, potential misuse and insurance cover. Renters expressed concerns about the condition of the vehicle and personal safety when interacting with strangers. These perceptions echo the conclusions from other studies showing that concerns about trust, reliability and liability can be strong barriers to participation in P2P car sharing and other ride-sharing schemes [20]. The ordinal regression analyses are consistent with this pattern, showing that perceived security and process complexity, especially regarding insurance and liability, are significantly associated with stated willingness to participate. This suggests that future pilot programs should prioritize the development and examination of trust and safety mechanisms to strengthen the reliability of the platforms.

Thirdly, sustainability and environmental awareness were found to be important, albeit secondary, motivators in both models. Individuals who placed more importance on environmental benefits showed a higher willingness to use P2P carsharing, which is consistent with existing literature that emphasizes the environmental argument for shared mobility models [39, 53]. However, the relatively low importance attached to "green" considerations shows that financial and trust-related aspects currently override environmental factors in consumer decision-making. This user perspective appears to be less closely aligned with Greece's stated environmental policy, which places strong emphasis on sustainability in the transport sector. A similar pattern appears across Europe, where studies indicate that environmental awareness and sensitivity among certain social groups, especially in the transport sector, remain weak [54, 55]. The European Environmental Agency has also made several efforts to better inform citizens about environmental and sustainability challenges and potential solutions [56].

The lack of statistically significant effects for most sociodemographic and travel behavior variables in our models contrasts with findings from more mature P2P carsharing markets. Studies from Florida, London, Paris, Madrid, and Tokyo report significant effects of gender, age, household composition, and income, often interpreted as reflecting lifestyle choices, risk tolerance, and familiarity with sharing practices [34, 57]. In these contexts, P2P carsharing is well established, allowing individual characteristics to meaningfully differentiate willingness to participate. In contrast, P2P carsharing in Athens is characterized by low awareness and limited platform presence. Under these conditions, attitudes related to cost, trust, and safety appear to play a more central role in shaping stated willingness than sociodemographic or travel-related characteristics.

This study contributes to the literature on peer-to-peer carsharing by providing empirical evidence from Greece, where P2P carsharing is still in its early stages, with limited platform presence and low public awareness. It also adds value by adopting a dual perspective, modeling willingness to participate both as a vehicle owner and as a renter within the same empirical framework, thus revealing differences in motivations and barriers between the two roles. Methodologically, the use of ordinal regression models, along with marginal effects and elasticity analysis, enables better interpretation of stated willingness and allows assessment of the relative influence of economic, environmental, and trust-related factors.

5.1 Limitations

Several limitations must be acknowledged, as they constrain the generalizability of the findings. First, the study used a non-random snowball sampling, which is efficient for recruiting participants but limits the external validity of the results [58, 59]. Additionally, respondents evaluated a hypothetical service, which further limits the external validity of the results. Willingness to participate may change once they interact with an operational P2P carsharing service, experience real procedures and costs, or observe actual trust and safety mechanisms.

The sample of 203 respondents may not fully represent the diverse socioeconomic and geographic characteristics of the entire Greek urban population. The recruitment process relied mainly on digital networks and academic or professional contacts, which likely favored younger adults, people with higher education, and residents of major urban centers. Older adults, people with limited digital literacy, lower-income groups, and residents of smaller towns are therefore less represented in the sample, and their mobility needs or attitudes toward P2P carsharing may not be fully captured. The increased participation of highly educated individuals could have led to an overestimation of the community's environmental awareness regarding the adoption of P2P services. Future research should consider the use of stratified or random sampling to increase the representativeness of the sample.

Furthermore, the statistical models cannot fully capture the complexity of personal mobility choices. Qualitative research through in-depth interviews and focus groups could uncover barriers related to cultural norms, peer influence or local infrastructural constraints that may be missed by quantitative tools.

5.2 Future research

Future research could enhance our understanding of P2P car sharing adoption in Greece by using data from existing platforms once they reach sufficient user volume. Future studies could analyze observed user behavior, platform user growth dynamics, and how users shape pricing policy based on supply and demand. This approach would enable assessment of whether willingness expressed in survey-based studies translates into actual participation, sustainable use, and balanced supply–demand dynamics.

Secondly, it would be useful to extend the investigation beyond mere willingness and examine frequency of use, modal shift and potential changes in car ownership rates (when pilot programs take place). If individuals using P2P carsharing significantly reduce their private car use, this could result in quantifiable environmental and transport improvements, strengthening the case for wider adoption.

Thirdly, a comparative study covering several countries or regions could shed light on cultural influences and infrastructural differences. Greece offers a unique setting with a relatively dense urban environment and inconsistent quality of public transportation, but comparable southern European countries as well as regions with a more advanced shared mobility ecosystem might show different patterns. Also, future research could focus on providing a better understanding of the emotional and cultural barriers affecting the adoption of P2P carsharing through qualitative methods that offer contextual insights beyond quantitative data.

Emerging transportation technologies, such as autonomous vehicles, are expected to enhance mobility by making it more integrated, equitable, and accessible [32, 48]. These technologies could shift P2P carsharing from a model where owners manually list and deliver their vehicles to one in which actions and processes are automated and tailored to user preferences, reducing the need for human involvement [15, 20, 60]. Advances in telematics and digital verification systems could further support trust, security, and adoption [15, 60]. However, the reliance on advanced technologies may create challenges for individuals with limited digital literacy and could even exclude them unless interfaces and actions are designed to accommodate diverse needs [15, 48]. Future research should therefore examine how emerging technologies will influence P2P carsharing, how these tools can be made more user-friendly and how both providers and renters will be affected by their integration.

Appendix A: survey questions

ID	Label	Measurement level
1	Do you use more than one mode of transport for your commute?	Ordinal
2	Does your daily commute involve walking more than 500 m (7 min)?	Ordinal
3	How many trips do you make in a typical day? (Assume that the outward and return journey corresponds to 2 trips)	Ordinal
4	What is the average total duration of a typical trip?	Ordinal
5	Do you consider the public transport network in your area to be adequate?	Ordinal
6	How important do you consider each of the following factors in mode choice? [Purpose of travel]	Ordinal
7	How important do you consider each of the following factors in mode choice? [Duration of travel]	Ordinal
8	How important do you consider each of the following factors in mode choice? [Comfort]	Ordinal
9	How important do you consider each of the following factors in mode choice? [Privacy]	Ordinal
10	How important do you consider each of the following factors in mode choice? [Safety]	Ordinal
11	How important do you consider each of the following factors in mode choice? [Flexibility]	Ordinal
12	How important do you consider each of the following factors in mode choice? [Environmentally friendly]	Ordinal
13	How important do you consider each of the following factors in mode choice? [Affordability]	Ordinal
14	How often do you use the following equipment or applications for your trips (trip planning/mode choice)? [Computer or laptop]	Ordinal
15	How often do you use the following equipment or applications for your trips (trip planning/mode choice)? [Smartphone or tablet]	Ordinal
16	How often do you use the following equipment or applications for your trips (trip planning/mode choice)? [Google Maps]	Ordinal
17	How often do you use the following equipment or applications for your trips (trip planning/mode choice)? [Telematics]	Ordinal
18	How often do you use the following equipment or applications for your trips (trip planning/mode choice)? [Taxi apps]	Ordinal
19	On a scale of 1 to 5 where: 1 means Not at all and 5: Very important. Please state how important are the following factors for using carsharing? [The possibility of renting different types of vehicles]	Ordinal
20	On a scale of 1 to 5 where: 1 means Not at all and 5: Very important. Please state how important are the following factors for using carsharing? [The possibility of renting electric vehicles]	Ordinal
21	On a scale of 1 to 5 where: 1 means Not at all and 5: Very important. Please state how important are the following factors for using carsharing? [Comfort of shared vehicle]	Ordinal
22	On a scale of 1 to 5 where: 1 means Not at all and 5: Very important. Please state how important are the following factors for using carsharing? [Free parking spaces for shared vehicles]	Ordinal
23	On a scale of 1 to 5 where: 1 means Not at all and 5: Very important. Please state how important are the following factors for using carsharing? [Availability of shared vehicles]	Ordinal

ID	Label	Measurement level
24	On a scale of 1 to 5 where: 1 means Not at all and 5: Very important. Please state how important are the following factors for using carsharing? [Ease of use]	Ordinal
25	On a scale of 1 to 5 where: 1 means Not at all and 5: Very important. Please state how important are the following factors for using carsharing? [Environmentally friendly]	Ordinal
26	On a scale of 1 to 5 where: 1 means Not at all and 5: Very important. Please state how important are the following factors for using carsharing? [Economic Benefit]	Ordinal
27	On a scale of 1 to 5 where: 1 means Not at all and 5: Very important. Please state how important are the following factors for using carsharing? [cleanliness inside the vehicle]	Ordinal
28	On a scale of 1 to 5, where 1 means Strongly Disagree and 5: Strongly Agree, please state your views on the following: [It is necessary to own a car]	Ordinal
29	On a scale of 1 to 5, where 1 means Strongly Disagree and 5: Strongly Agree, please state your views on the following: [Owning a car is a matter of social status]	Ordinal
30	On a scale of 1 to 5, where 1 means Strongly Disagree and 5: Strongly Agree, please state your views on the following: [Owning a car offers freedom and flexibility]	Ordinal
31	On a scale of 1 to 5, where 1 means Strongly Disagree and 5: Strongly Agree, please state your views on the following: [I know exactly what the annual cost of owning & maintaining a car is]	Ordinal
32	What would motivate you as a car owner to participate in a carsharing scheme between private individuals? (1 = not important, 5 = very important) [Economic Benefit]	Ordinal
33	What would motivate you as a car owner to participate in a carsharing scheme between private individuals? (1 = not important, 5 = very important) [Environmental Concerns]	Ordinal
34	What are your main concerns about your participation? (1 = not important, 5 = very important) [Lack of trust in other users]	Ordinal
35	What are your main concerns about your participation? (1 = not important, 5 = very important) [Vehicle wear and tear]	Ordinal
36	What are your main concerns about your participation? (1 = not important, 5 = very important) [Insurance and liability]	Ordinal
37	What are your main concerns about your participation? (1 = not important, 5 = very important) [Maintenance and mechanical breakdowns]	Ordinal
38	What are your main concerns about your participation? (1 = not important, 5 = very important) [Cleanliness of the vehicle on return]	Ordinal
39	What are your main concerns about your participation? (1 = not important, 5 = very important) [Privacy & data protection issues]	Ordinal
40	How important do you consider the following factors for renting your own car to other users? (1 = not at all important, 5 = very important) [Provision of full vehicle insurance by the marketplace]	Ordinal
41	How important do you consider the following factors for renting your own car to other users? (1 = not at all important, 5 = very important) [Determining the availability of the vehicle by me]	Ordinal
42	How important do you consider the following factors for renting your own car to other users? (1 = not at all important, 5 = very important) [Approval of the renter by me]	Ordinal
43	How important do you consider the following factors for renting your own car to other users? (1 = not at all important, 5 = very important) [Setting the rental price by me]	Ordinal
44	How important do you consider the following factors for renting your own car to other users? (1 = not at all important, 5 = very important) [Automated vehicle delivery/pick-up]	Ordinal
45	How willing would you be to rent your own car to others via a P2P car sharing service?	Ordinal
46	What would motivate you to participate as a renter in a P2P carsharing scheme? (1 = not important, 5 = very important) [Immediate availability of vehicles]	Ordinal
47	What would motivate you to participate as a renter in a carsharing scheme between private individuals? (1 = not important, 5 = very important) [Convenience at the point of pick-up & delivery of the vehicle]	Ordinal
48	What would motivate you to participate as a renter in a P2P carsharing scheme ? (1 = not important, 5 = very important) [Variety of vehicles (different makes & models)]	Ordinal
49	What would motivate you to participate as a renter in a P2P carsharing scheme? (1 = not at all important, 5 = very important) [More economical than traditional vehicle rental]	Ordinal
50	What would motivate you to participate as a renter in a P2P carsharing scheme? (1 = not important, 5 = very important) [Environmental concerns]	Ordinal

ID	Label	Measurement level
51	What are your main concerns about your participation? (1 = not important, 5 = very important) [Lack of vehicle availability]	Ordinal
52	What are your main concerns about your participation? (1 = not important, 5 = very important) [Feeling unsafe about the condition of the vehicle]	Ordinal
53	What are your main concerns about your participation? (1 = not important, 5 = very important) [Feeling unsafe towards the owner of the vehicle]	Ordinal
54	What are your main concerns about your participation? (1 = not important, 5 = very important) [Lack of familiarity with the service]	Ordinal
55	What are your main concerns about your participation? (1 = not important, 5 = very important) [Complexity of the hiring process]	Ordinal
56	How important do you consider the following factors to rent a car from another private person? (1 = not at all important, 5 = very important) [Provision of full vehicle insurance]	Ordinal
57	How important do you consider the following factors to rent a car from another private person? (1 = not at all important, 5 = very important) [Transparent terms of use]	Ordinal
58	How important do you consider the following factors to rent a car from another private person? (1 = not at all important, 5 = very important) [Automated delivery/pick-up of the vehicle (without the physical presence of the owner)]	Ordinal
59	How willing would you be to rent a car from a private individual via a P2P car sharing service?	Ordinal
60	What age group do you belong to?	Nominal
61	What category does your monthly net family income fall into?	Nominal
62	How many members does your household consist of? (including yourself)	Nominal
63	How many vehicles are there in your household? [Car]	Scale
64	How many vehicles are there in your household? [EV]	Scale
65	How many vehicles are there in your household? [Motorcycle]	Scale
66	How many vehicles are there in your household? [Bicycle]	Scale
67	How many vehicles are there in your household? [Scooter]	Scale
68	Do you hold a driving licence?	Nominal
69	How often do you have access to any of your household vehicles?	Ordinal
70	Have you ever heard of P2P carsharing before?	Nominal
71	What is the main purpose of your daily commute?	Nominal
72	What is the main mode of transport you use?	Nominal
74	What is your occupation?	Nominal
75	What is your previous level of study?	Nominal
76	How would you describe the area where you live?	Nominal

Appendix B: survey's variable values and responses

Value		Label	%
Main trip purpose	1	Household Obligations	8.4%
	2	Leisure activities	4.4%
	3	Other	0.5%
	4	Shopping	4.9%
	5	Work/ Education	81.8%
Main transport mode	1	Motorcycle	5.4%
	2	Private vehicle as driver	60.1%
	3	Private vehicle as passenger	3%
	4	Public Transport	28.1%
	5	Walking	3.4%
Gender	1	Man	46.8%
	2	Woman	53.2%
Occupation	1	Do not wish to answer	0.5%
	2	Employee	63%
	3	Household	0.5%

Value		Label	%
	4	Retired	4.4%
	5	Self employed	19.7%
	6	Student	9.9%
	7	Unemployed	1%
	8	Other	1%
Education	1	Bachelor	45.8%
	2	Compulsory Education	2%
	3	Do not wish to answer	0.5%
	4	High School Graduate	13.3%
	5	Master	27.1%
	6	PhD	4.9%
	7	Technical College	6.4%
Monthly family income	1	< 1000 €	11.3%
	2	1001–1500 €	19.2%
	3	1501–2500 €	28.1%
	4	2501–3750 €	14.8%
	5	3751–5000 €	8.9%
	6	> 5000€	5.9%
	7	Do not wish to answer	11.8%
Driving license	1	No	12.3%
	2	Yes	87.7%
Type of area	1	Rural	3.9%
	2	Suburban	49.3%
	3	Urban	46.8%

Appendix C: model 1- willingness to participate as car owner: marginal effects and elasticities

Marginal effects

Factor	AME	SE	z	p	lower	upper
What would motivate you as a car owner to participate in a carsharing scheme between private individuals? [Economic Benefit] (Not at all important to Unimportant)	0.7087	0.097	7.3042	0	0.5185	0.8988
What would motivate you as a car owner to participate in a carsharing scheme between private individuals? [Economic Benefit] (Unimportant to neutral)	0.1867	0.0949	1.9671	0.0492	0.0007	0.3727
What would motivate you as a car owner to participate in a carsharing scheme between private individuals? [Economic Benefit] (Neutral to important)	0.2271	0.0869	2.6137	0.009	0.0568	0.3974
What would motivate you as a car owner to participate in a carsharing scheme between private individuals? [Economic Benefit] (Important to very important)	0.085	0.0491	1.731	0.0835	-0.0112	0.1813
What would motivate you as a car owner to participate in a carsharing scheme between private individuals? [Environmental Concerns] (Not at all important to Unimportant)	0.1291	0.0837	1.5436	0.1227	-0.0348	0.2931

Factor	AME	SE	z	p	lower	upper
What would motivate you as a car owner to participate in a carsharing scheme between private individuals? [Environmental Concerns] (Unimportant to neutral)	0.1194	0.0689	1.7318	0.0833	-0.0157	0.2544
What would motivate you as a car owner to participate in a carsharing scheme between private individuals? [Environmental Concerns] (Neutral to important)	0.1196	0.063	1.8977	0.0577	-0.0039	0.2431
What would motivate you as a car owner to participate in a carsharing scheme between private individuals? [Environmental Concerns] (Important to very important)	0.0599	0.056	1.0706	0.2843	-0.0498	0.1697
What are your main concerns about your participation? [Vehicle wear and tear] (Not at all important to Unimportant)	-0.2309	0.1241	-1.8601	0.0629	-0.4742	0.0124
What are your main concerns about your participation? [Vehicle wear and tear] (Unimportant to neutral)	-0.2533	0.0579	-4.3758	0	-0.3667	-0.1398
What are your main concerns about your participation? [Vehicle wear and tear] (Neutral to important)	-0.0908	0.0648	-1.4021	0.1609	-0.2177	0.0361
What are your main concerns about your participation? [Vehicle wear and tear] (Important to very important)	-0.1619	0.0471	-3.438	0.0006	-0.2543	-0.0696
How important do you consider the following factors for renting your own car to other users? [Approval of the renter by me] (Not at all important to Unimportant)	-0.1097	0.1169	-0.9381	0.3482	-0.3388	0.1195
How important do you consider the following factors for renting your own car to other users? [Approval of the renter by me] (Unimportant to neutral)	-0.2144	0.0649	-3.3041	0.001	-0.3416	-0.0872
How important do you consider the following factors for renting your own car to other users? [Approval of the renter by me] (Neutral to important)	-0.1308	0.0535	-2.4444	0.0145	-0.2357	-0.0259
How important do you consider the following factors for renting your own car to other users? [Approval of the renter by me] (Important to very important)	-0.1313	0.0474	-2.7706	0.0056	-0.2241	-0.0384

Elasticities

Factor	Elasticity
What would motivate you as a car owner to participate in a carsharing scheme between private individuals? [Economic Benefit] (Not at all important to Unimportant)	0.01385337
What would motivate you as a car owner to participate in a carsharing scheme between private individuals? [Economic Benefit] (Unimportant to neutral)	0.28139922
What would motivate you as a car owner to participate in a carsharing scheme between private individuals? [Economic Benefit] (Neutral to important)	0.22623989
What would motivate you as a car owner to participate in a carsharing scheme between private individuals? [Economic Benefit] (Important to very important)	0.51829965
What would motivate you as a car owner to participate in a carsharing scheme between private individuals? [Environmental Concerns] (Not at all important to Unimportant)	0.34967215
What would motivate you as a car owner to participate in a carsharing scheme between private individuals? [Environmental Concerns] (Unimportant to neutral)	0.37350949

Factor	Elasticity
What would motivate you as a car owner to participate in a carsharing scheme between private individuals? [Environmental Concerns] (Neutral to important)	0.37291899
What would motivate you as a car owner to participate in a carsharing scheme between private individuals? [Environmental Concerns] (Important to very important)	0.57919539
What are your main concerns about your participation? [Vehicle wear and tear] (Not at all important to Unimportant)	6.95530978
What are your main concerns about your participation? [Vehicle wear and tear] (Unimportant to neutral)	9.85921346
What are your main concerns about your participation? [Vehicle wear and tear] (Neutral to important)	1.78089232
What are your main concerns about your participation? [Vehicle wear and tear] (Important to very important)	3.16307313
How important do you consider the following factors for renting your own car to other users? [Approval of the renter by me] (Not at all important to Unimportant)	2.15357046
How important do you consider the following factors for renting your own car to other users? [Approval of the renter by me] (Unimportant to neutral)	6.68265939
How important do you consider the following factors for renting your own car to other users? [Approval of the renter by me] (Neutral to important)	2.58362195
How important do you consider the following factors for renting your own car to other users? [Approval of the renter by me] (Important to very important)	2.59342189

Appendix D: model 2—willingness to participate as car owner: marginal effects and elasticities

Marginal effects

Factor	AME	SE	z	p	lower	upper
Please state how important are the following factors for using carsharing? [Economic Benefit] (Not at all important to Unimportant)	0.3974	0.1831	2.1702	0.03	0.0385	0.7563
Please state how important are the following factors for using carsharing? [Economic Benefit] (Unimportant to Neutral)	-0.0116	0.039	-0.2968	0.7666	-0.0881	0.0649
Please state how important are the following factors for using carsharing? [Economic Benefit] (Neutral to Important)	0.0077	0.0298	0.2597	0.7951	-0.0507	0.0662
Please state how important are the following factors for using carsharing? [Economic Benefit] (Important to Very Important)	-0.0142	0.0198	-0.7165	0.4737	-0.053	0.0246
What would motivate you to participate as a renter in a P2P carsharing scheme? [Immediate availability of vehicles] (Not at all important to Unimportant)	0.3549	0.1835	1.9338	0.0531	-0.0048	0.7145
What would motivate you to participate as a renter in a P2P carsharing scheme? [Immediate availability of vehicles] (Unimportant to Neutral)	0.1696	0.0912	1.8599	0.0629	-0.0091	0.3484
What would motivate you to participate as a renter in a P2P carsharing scheme? [Immediate availability of vehicles] (Neutral to Important)	0.1001	0.0482	2.0783	0.0377	0.0057	0.1945
What would motivate you to participate as a renter in a P2P carsharing scheme? [Immediate availability of vehicles] (Important to Very Important)	0.0262	0.0209	1.2579	0.2084	-0.0147	0.0672

Factor	AME	SE	z	p	lower	upper
What would motivate you to participate as a renter in a P2P carsharing scheme? [Environmental concerns] (Not at all important to Unimportant)	0.0076	0.0331	0.2298	0.8182	-0.0572	0.0724
What would motivate you to participate as a renter in a P2P carsharing scheme? [Environmental concerns] (Unimportant to Neutral)	0.1019	0.0519	1.9633	0.0496	0.0002	0.2035
What would motivate you to participate as a renter in a P2P carsharing scheme? [Environmental concerns] (Neutral to Important)	0.0291	0.027	1.0785	0.2808	-0.0238	0.082
What would motivate you to participate as a renter in a P2P carsharing scheme? [Environmental concerns] (Important to Very Important)	0.0221	0.0277	0.7993	0.4241	-0.0322	0.0764
What are your main concerns about your participation? [Feeling unsafe towards the owner of the vehicle] (Not at all important to Unimportant)	-0.0293	0.074	-0.3964	0.6918	-0.1743	0.1157

Elasticities

Factor	Elasticity
Please state how important are the following factors for using carsharing? [Economic Benefit] (Not at all important to Unimportant)	0.06040085
Please state how important are the following factors for using carsharing? [Economic Benefit] (Unimportant to Neutral)	1.1960042
Please state how important are the following factors for using carsharing? [Economic Benefit] (Neutral to Important)	0.89674878
Please state how important are the following factors for using carsharing? [Economic Benefit] (Important to Very Important)	1.24909428
What would motivate you to participate as a renter in a P2P carsharing scheme? [Immediate availability of vehicles] (Not at all important to Unimportant)	0.04744974
What would motivate you to participate as a renter in a P2P carsharing scheme? [Immediate availability of vehicles] (Unimportant to Neutral)	0.14046268
What would motivate you to participate as a renter in a P2P carsharing scheme? [Immediate availability of vehicles] (Neutral to Important)	0.24739787
What would motivate you to participate as a renter in a P2P carsharing scheme? [Immediate availability of vehicles] (Important to Very Important)	0.60576907
What would motivate you to participate as a renter in a P2P carsharing scheme? [Environmental concerns] (Not at all important to Unimportant)	0.85988939
What would motivate you to participate as a renter in a P2P carsharing scheme? [Environmental concerns] (Unimportant to Neutral)	0.25316247
What would motivate you to participate as a renter in a P2P carsharing scheme? [Environmental concerns] (Neutral to Important)	0.59743675
What would motivate you to participate as a renter in a P2P carsharing scheme? [Environmental concerns] (Important to Very Important)	0.66649132
What are your main concerns about your participation? [Feeling unsafe towards the owner of the vehicle] (Not at all important to Unimportant)	1.37909839
What are your main concerns about your participation? [Feeling unsafe towards the owner of the vehicle] (Unimportant to Neutral)	3.58778452
What are your main concerns about your participation? [Feeling unsafe towards the owner of the vehicle] (Neutral to Important)	3.3946881
What are your main concerns about your participation? [Feeling unsafe towards the owner of the vehicle] (Important to Very Important)	2.29795632
What are your main concerns about your participation? [Complexity of the hiring process] (Not at all important to Unimportant)	7.27095366

Factor	Elasticity
What are your main concerns about your participation? [Complexity of the hiring process] (Unimportant to Neutral)	3.99032206
What are your main concerns about your participation? [Complexity of the hiring process] (Neutral to Important)	2.79165572
What are your main concerns about your participation? [Complexity of the hiring process] (Important to Very Important)	3.10762249

Author contributions

C.K. and A.D. Conceptualization, C.K., A.D. and L.M. Methodology, C.K., E.A., and A.D. Formal analysis, A.D. and E.A. Writing—Original Draft, C.K. and L.M. Writing—Review & Editing.

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Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval

This study has been conducted in strict accordance with the General Data Protection Regulation (GDPR) (EU) 2016/679 and was approved by the Ethics Committee of Research of National Technical University of Athens (NTUA).

Consent to participate

Informed consent was obtained from all individual participants included in the study.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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